

SUPPORTING INVESTMENT MEMO — CONFIDENTIAL

TWB Designs — the demilitarization and global secure destruction platform

Companion to the investor deck (TWB_Investor_Deck_Sep2026.pptx) and the financial model (TWB_Financial_Model.xlsx). Prepared September 2026.

Purpose: expand the deck's narrative for investor review and internal alignment — the current foundation, the market opportunity, the platform thesis, the strategic roadmap, the operating and financial model, risks and mitigants, and the investment case — with every figure traceable to a named source, a company record or a labelled model assumption.

This memo is the only document in the package that presents the current business's actual financial results (Section 2). It should travel with the deck.

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1. Purpose, scope and how to read the numbers

This memo provides the sourcing, assumptions and reasoning behind two companion deliverables: the investor deck and the financial model. It is organised so that any number in the deck can be traced to a real financial statement, a cited external source, or an explicitly labelled modelling assumption. It replaces the August 2026 supporting memo in full.

Three kinds of numbers appear throughout the package, and each is labelled:

- VERIFIED — cited to a named primary source with a date and a live link (Sources tab of the model; Sources slides of the deck).
- COMPANY-PROVIDED — from TWB's own records, product documentation or management (marked "company"). Third-party verification of the recovery-rate and emissions-standard claims has not yet been published; until it is, they are presented as operating figures.
- PROJECTION / OUR CONSTRUCTION — the market construction and the ten-year plan, built in the model from labelled assumptions. These are planning cases, not forecasts of results.

Where a claim in the August package could not be verified it has been removed, softened or marked as an assumption. The fact-check log (model tab and standalone document) lists every claim, its status, the support found and what changed. The deck is deliberately forward-looking and does not present current revenue or EBITDA; those figures are disclosed here in Section 2 as the diligence baseline.

All figures are in US dollars unless noted. Canadian-dollar actuals are converted at Bank of Canada annual average rates (2023: 1.3497; 2024: 1.3698; 2025: 1.3978).

2. Where we are today — the foundation

2.1 The business

TWB Designs Inc. (Hamilton, Ontario) owns and operates eMACS, a trailer-deployed, two-stage closed-chamber combustion system that destroys small-arms ammunition to .50 calibre — plus seized narcotics and other seized property for law-enforcement customers — on site, same day, inside the owner's perimeter, and recovers the metal. The system was developed in a joint venture with the Royal Canadian Mounted Police and has been in use since 2001. Its sister company, Canadian Ammunition Disposal Services Inc. (CADS), performs the destruction service for Canadian police agencies using the equipment. The acquisition of both companies by the current ownership group closed in 2026, and a US entity has been established.

Deployments and relationships (company-provided unless noted): Canadian federal, provincial and municipal policing (RCMP units, Sûreté du Québec, Calgary and Winnipeg police services per the CADS customer list); US county law enforcement, including the Franklin County (Ohio) Sheriff's Office and Hilliard Division of Police take-back events (1,630 lb / ~54,000 rounds collected on 21 March 2026 — event verified; TWB's role as disposal vendor is company-provided); a multi-country sales agreement with Military Equipment Denmark A/S covering Denmark, Norway, Sweden, Finland, Latvia, Lithuania and Estonia (announced 2013); and deployments with the Australian Army, the New Zealand Army and the Royal Saudi Land Forces (company-provided; references under NDA). eMACS was a finalist in the 2011 Law Enforcement Technology Innovation Awards presented at the IACP annual conference.

2.2 The technology

Company-published specifications: ~850°F primary chamber and ~1,750°F afterburner; up to ~100 lb (45.4 kg) per one-hour cycle and ~1,000 lb per unit per day; tested to .50 calibre; 97% of recyclable material recovered per cycle; the company states the system meets or exceeds California EPA, NATO and EU emissions standards, holds an Ontario Environmental Compliance Approval (MECP) and is TSSA field-approved. Trailer: 11,500 lb, Class V hitch, armoured 3/8-inch shell, remote start/stop. The original patents — US 6,758,151 (2004) and CA 2,402,939 (2008), "Remotely activated armored incinerator with gas emission control", inventors Braithwaite and Ivany, assigned to the Crown/RCMP — have expired; the platform's protection is 25 years of accumulated process know-how, references and approvals rather than live patents.

A citable regulatory fact: 40 CFR 260.10 defines open burning as combustion without control of combustion air, containment in an enclosed device, and control of emissions. A closed two-stage chamber is categorically not open burning. Environmental counsel should settle whether "two-stage combustion" wording places the unit in the incinerator definition rather than a Subpart X miscellaneous unit before any further regulatory claim is locked.

2.3 The financial baseline (company-provided; compiled, unaudited)

Combined figures for TWB Designs Inc. and CADS from compiled financial statements (Steve Herman Accounting), FY2023–FY2025. These are compilation engagements — no assurance is provided. Diligence requires bank statements, HST filings, receivables ageing and customer contracts.

CAD	FY2023	FY2024	FY2025
CADS revenue (destruction services)	901,305	847,761	720,044
TWB Designs revenue (equipment sales)	64,425	863,500	142,812
Combined revenue	965,730	1,711,261	862,856
Combined gross profit	759,474	760,306	569,099
Combined EBITDA (approx.)	~419,000	~492,000	~235,000
Combined net income	337,094	589,471	125,174
Combined revenue, USD (BoC annual average)	\$0.72M	\$1.25M	\$0.62M

Balance sheet at 31 December 2025 (CAD, combined): cash 199,098; total assets 628,018; total liabilities 47,689; book equity 580,329; no interest-bearing debt (the only loan, a CAD 40K CEBA facility at CADS, was repaid in 2023). Notes: CADS pays TWB a CAD 60K annual intercompany fee, so the simple sum overstates consolidated revenue by that amount; FY2024 net income included a one-time CAD 265,874 foreign-exchange gain; equipment sales are lumpy (one unit in FY2024); services revenue declined about 20% across 2023–2025.

How the model treats this: FY2025 actuals are the anchor. The FY2026 starting point is a marked placeholder (Baseline tab, yellow cell) that defaults to the FY2025 actual until management supplies a 2026 estimate; the Year-0 fleet revenue never starts below the last actual. The August package modelled a \$3.5M Year-0 base; this package does not.

2.4 What the foundation gives the platform

Beyond revenue, the foothold supplies three things the platform is built from: an operating doctrine — same day, on site, the owner's own people, zero transfers, recovered not released, nobody standing over live ordnance — that every acquired and developed capability will run on; a quarter-century of destruction records, references and approvals; and the certification and chain-of-custody data that no incumbent in the category is collecting. The custody wedge is the sharpest proof point eMACS carries into any deal: the fixed-site incumbents still move the material.

3. Why the opportunity is bigger than the business

3.1 The rearmament and the liability it creates

World military expenditure reached \$2,887 billion in 2025, an all-time record and the eleventh consecutive annual rise; European spending doubled over 2016–25 (SIPRI, April 2026). There were 65 state-based armed conflicts in 2025, the most since records began in 1946, eight of them between states (UCDP/JPR, June 2026). NATO has identified up to \$145 billion in shared munitions and air-defence requirements and reports Allied production up roughly sixfold (NATO Annual Report 2025 briefing, March 2026). Rheinmetall states a 155mm capacity of at least 1.1 million rounds a year by 2027, with company statements reported as high as 1.5 million; the US Army's 100,000-rounds-a-month goal remains unmet at roughly 36,000 (DoD IG, July 2026).

Ammunition is the only weapon that must be destroyed to be retired: propellants destabilise, primers corrode, calibres are orphaned when the gun changes. Every round produced today is a future destruction obligation. The US demilitarization account held 296,534 tons at 4 December 2025 across more than 7,000 DODICs (FY2027 Army Procurement of Ammunition justification book; PD Demil, October 2025). Two facts about that queue should be stated together: it has fallen 37% since end-FY2016 as the Army works it down at a funded rate of roughly 30,000 tons a year, while missile components have grown from 6.5% to 16.8% of the account; and the Army's own October 2025 forecast has generations rising again to about 361,000 short tons by FY2030 as surge production ages into the stockpile.

3.2 Destruction never industrialised

The incumbent methods are a pit, a truck and a pair of hands. Open burning and open detonation cost roughly \$750 a ton (Army estimate cited by the National Academies, 2019) and are being regulated out: EPA's proposed rule (89 FR 19952, March 2024) covers the 67 facilities with operating OB/OD units, with a final rule projected for late 2026 — it is not yet final. Contained burn and detonation is estimated by the Army at \$2,000–20,000 a ton and the average conventional demil cost at \$2,890 a ton (2017 DoD report cited by the National Academies). The alternative to a pit is a truck to a permitted depot, with every mile a diversion window and every transfer a custody break. The default when nothing else is on site is human hands: on 18 July 2025 three Los Angeles County Sheriff's detectives were killed at the Biscailuz Center while handling seized ordnance.

3.3 The flip — critical minerals

The Final 2025 US Critical Minerals List (90 FR 50494, 7 November 2025) added copper and lead; antimony has been on the list since 2022. Antimony prices rose from an average of \$5.49/lb in 2023 to \$10.24/lb in 2024 and about \$25/lb in 2025, peaking near \$27.50/lb (about \$60,000 a ton) in mid-2025 after

China restricted and then banned exports to the United States (USGS Mineral Commodity Summaries 2026) — roughly a fivefold move from early 2024 to the peak. The US Export–Import Bank approved a \$2.9 billion loan for a domestic antimony source in May 2026 (Perpetua Resources), and GAO-26-108687 (July 2026) frames domestic recycling feedstock as a critical-minerals priority. The August deck's "2,600%" and "\$1B Pentagon commitment" figures could not be supported and have been replaced with these primary-source figures.

One rule governs how recovery economics are underwritten: in US DoD demil services the Army sells recyclable material itself and retains the proceeds (10 USC 7690); title passes only where a licensed manufacturer buys the items outright (10 USC 7687). Recovery is therefore modelled only on law-enforcement, commercial and allied work where the material is the customer's or the platform's to sell — never on a metals margin from DoD demil services.

3.4 The market — our construction

No analyst publishes a munitions end-of-life market. The Market tab builds it layer by layer, each labelled sourced or assumed:

Layer (USD M / yr, 2026)	Value	Basis
US Army conventional munitions demil line (FY2026)	155	Sourced — FY2027 J-book (\$161M requested FY2027)
Other US DoD / federal destruction & demil services	150	Assumed placeholder — no consolidated public figure
Allied national demil & stockpile-destruction budgets	450	Assumed — ~3x the US Army line across ~30 allied MoDs
Law-enforcement, civil & commercial secure destruction (NA + EU)	100	Assumed — no published aggregate
EOD services share (30%) of the \$8.55bn EOD market	2,565	Sourced size (Global Growth Insights, 3.03% CAGR); service-fit factor assumed
Humanitarian mine action funding	1,070	Sourced — Landmine Monitor 2025 (\$1.07bn in 2024)
Sovereign end-of-life facility engineering	500	Assumed placeholder
Recovered materials sold into secondary markets	250	Assumed — ~5% of destruction-services layers
TAM — forming annual spend	5,240	Blended growth ~3%/yr
Episodic layer — Ukraine explosive-hazard management, 2026–2035	27,600 (total)	Sourced — RDNA5, Feb 2026 (supersedes RDNA4 \$29.8bn)
SAM — North America + Europe, 60% geographic × 70% access	2,201	Assumed factors
SOM — model Year-5 revenue (2031)	137	5.4% of Year-5 SAM; 11.6% of Year-10 SAM at Year 10

The August package's \$15bn TAM — which included a chemical-weapons segment and an unsourced \$2.5bn ammunition-demil line — is withdrawn. Discipline: the platform is never scored on tons per year; the metric is rounds destroyed without leaving the compound, and net cost of destruction is pitched rather than "cheaper" in the gross.

3.5 Proof the buyer is spending

Verified demand signals: the Army funds a dedicated conventional demil line every year (\$155.1M FY2026, \$161.2M FY2027 requested, 31,475 tons funded); PD Demil plans a competitive ~\$30M-a-year Conventional Ammunition Demil IDIQ for 2026–2030, with the prior \$110.1M three-awardee contract (GD-OTS, EnviroSafe Demil, Expal USA) having ended 20 May 2026 — the successor award had not been evidenced publicly as of 1 September 2026; Day & Zimmermann received a \$2,301,227,487 contract on 9 June 2026 for operation, maintenance and modernisation of Hawthorne Army Depot including demil of specified ammunition families, through December 2046; the UN ran two procurements in 2025 for containerised mobile small-arms ammunition destruction (RFIUNPD23632; EOIUNPD23750, ≥40 kg/hr, deployable to eight missions); PSPC issued W8486-217446/B for DND ammunition and explosives demilitarization and disposal services; and the UK MoD published a £23M EOD and Search supportability framework pipeline notice (August 2026). The August deck's "\$808.8M IDIQ", "\$3bn across 2026–27 demil awards" and "small-business set-aside for new entrants" claims could not be verified and have been withdrawn.

4. What the company can become — the platform thesis

The strangest fact in this market is that almost everything needed to unmake the world's munitions already exists — scattered. Artillery demilitarization houses, energetics and metals-recovery specialists, destruction-plant engineers and secure-destruction operators: dozens of firms, most founded in the 1980s and 1990s, most still founder- or family-held, each holding one capability, one site and one generation of tribal knowledge. What has never existed is the company: no consolidator at the founder scale, no single counterparty a government can hand the whole problem to. Strategics are buying the large end — Rheinmetall–Expal (2023, ~€1.2bn, ~3.4x 2021 revenue), Diehl–Tauber (October 2025, ~€100M revenue), Diehl–EST Energetics (July 2026), Arca Defence–Esplodenti Sabino (October 2025), and Cadre Holdings' 18 acquisitions — which validates the category while leaving the founder-scale layer unassembled.

The platform is assembled through three engines run in parallel — Acquire, Partner, Develop — into six service lines sold by one counterparty: deployable custody destruction (operating today); large-calibre and artillery demilitarization; materials recovery and reclamation; plant and equipment engineering; automated and unmanned processing; and certification and chain-of-custody data. Governments already buy every line separately from single-capability vendors; the platform sells the lifecycle. Per the brief, individual capability lines are not assigned to a delivery mode in the deck; the model's acquisition programme names capability categories only, and the ranked, contact-ready target shortlist is held for diligence.

The end state: when an allied nation decides ammunition must cease to exist, the call comes to the platform — and the property never leaves their custody on the way. Sovereignty is the product: Canada is a DFARS 225.872 qualifying country, so Canadian end products are exempt from Buy American restrictions in US defence acquisition; a US entity is established; the positioning is US-ready and allied-forward. The Canadian entity does not claim US small-business status, an ATF FEL or DDTC registration.

5. How we get there — roadmap and operating model

5.1 Technology roadmap

Each stage bolts onto the mobile closed-chamber platform rather than replacing it. Today: small arms through .50 calibre with metals recovery. Year 1 (2027): large-calibre and bombs via an energetics-dissolution front end licensed from a US university (in discussion; name withheld), with the thermal unit treating residues. Year 2 (2028): propellant and rocket-motor demil, metals-discrimination sensing with a university partner (placeholder pending confirmation), and a tactical-equipment line for EOD and mine-action teams. Year 3 (2029): robotic feed and cutting for large-calibre and UXO, enabling unattended operation and post-conflict tenders. Year 5+ (2031): containerised and fixed-site formats and the sovereign design-build-operate-certify offer. Enabling tracks run continuously: certification and permitting (EPA/MECP, NATO STANAG, export-control classification), metals-recovery scale-up, a residuals pathway, data and traceability, and training and services. Chemical-weapons demilitarization, carried as Year-10 optionality in August, is withdrawn as OPCW-controlled, shrinking and off-strategy.

5.2 Geography

Canada and the United States today; Europe and NATO from 2028, accelerated by the agent network already in place across seven Nordic and Baltic countries and by the European acquisition in the programme; post-conflict programmes from 2029, entered with an established allied track record; Middle East and allied partners in later years, building on existing Australian, New Zealand and Saudi deployments once delivery at scale is demonstrated in two other regions.

5.3 The phase ladder and what each phase funds

Phase	Timing	What happens	Capital required (base)
1 — Operate the foothold	2026–27	Fleet build-out, platform team, certification layer, first allied programmes; first acquisitions negotiated	\$2.0M
2 — Assemble the core	2028–29	First capability acquisitions (large-calibre demil, metals recovery, plant engineering); integration; first robotics and university partnerships	\$41.5M
3 — The sovereign full-service offer	2030–31	Secure-destruction and European acquisitions; sovereign programme mobilisation; working capital	\$5.8M
Later years	2032–36	Remaining acquisition; growth increasingly self-funded from platform cash flow	\$0.0M

5.4 The operating model — how the plan is built

Organic lines (Organic Build tab). Custody destruction services are fleet-driven from 2027: units in service (2 today — placeholder to confirm — plus 1 in 2027, 2 in 2028 and 3 a year thereafter in the base case) times revenue per fully utilised unit-year (\$0.50M base, escalating 3% a year — an assumption; FY2025 services revenue was ~\$0.52M across the existing fleet, so the plan assumes utilisation rises under the US entity and allied demand), with new units earning half a year in their first year. Year 0 (2026) is the Baseline placeholder, which defaults to the FY2025 actual. Equipment sales via agents: units sold (1–3 a year from 2027) at \$0.65M, anchored to the FY2024 sale. Allied and sovereign programmes from 2030 at \$10M a

year each (assumption/placeholder), reaching eight by 2036. Post-conflict programmes from 2029 at \$5M per programme-year. Metals recovery at 10% of custody-services revenue (down from the August 14% placeholder), on law-enforcement, commercial and allied work only.

Acquisition programme (Acquisitions tab). Six capability acquisitions in the base case, priced deal by deal at EBITDA multiples bounded by lower-middle-market data (GF Data 2025: sub-\$10M TEV 5.5x, \$10–25M 6.4x; Pepperdine 2025: 4.0–8.3x) rather than the August 1.2x-revenue assumption, which every disclosed sector transaction 2023–26 exceeded (Cadre–Carr's ~1.5x, Cadre–ICOR ~2.3x, Cadre–Alpha Safety ~2.4x, Rheinmetall–Expal ~3.4x revenue). Post-close growth 12% a year (August: 15%); margin uplift of four points phased over three years; integration cost 10% of price. A pricing switch re-prices every deal at 2.0x revenue as a stress case.

Capability category (base close year)	Entry revenue	EBITDA margin	Multiple	Price
Large-calibre & artillery demilitarization (2028)	\$18.0M	18%	5.5x	\$17.8M
Metals recovery & reclamation (2028)	\$15.0M	12%	5.0x	\$9.0M
Plant & equipment engineering (2029)	\$12.0M	15%	5.5x	\$9.9M
Secure destruction — military & conflict customers (2030)	\$10.0M	20%	6.0x	\$12.0M
European demilitarization operating business (2031)	\$20.0M	18%	6.0x	\$21.6M
Energetics / second recovery capability (2033)	\$15.0M	15%	5.5x	\$12.4M
Total programme (base)	\$90M		0.92x revenue	\$82.7M
Stress — every deal at 2.0x revenue			2.0x	\$180M

Costs and capital (P&L & Capital tab). Contribution margins: custody services 30%, equipment 25%, sovereign programmes 15%, post-conflict 15%, metals 50% (all assumptions). Corporate platform cost was originally built bottom-up on a schedule from \$1.2M (2026) to \$11M (2036) plus 2% of acquired revenue; that bottom-up schedule still governs Year 0, but from Year 1 onward corporate platform cost is instead solved residually against a stated blended Adjusted EBITDA margin target — 30% by Year 5, growing to 35% by Year 10 (Tom, 3 Sep 2026; see the model's Inputs tab) — so consolidated margin hits that path exactly rather than falling out of the original cost build. Development and partnership programmes are expensed at \$0.3–3.0M a year. Cash taxes at a simplified 26% on positive EBITDA less D&A; fleet capex at \$0.40M per unit built (placeholder); maintenance capex 2.5% of revenue; working capital 12% of incremental revenue. Equity is drawn only when cash would otherwise fall below zero, so the capital requirement is the model's own output, on a fully equity-funded basis with no acquisition debt or seller paper.

6. The financial plan

6.1 Base case by year

USD M	Y0 · 2026	Y2 · 2028	Y4 · 2030	Y5 · 2031	Y6 · 2032	Y8 · 2034	Y10 · 2036
Organic revenue	\$0.6M	\$3.0M	\$24.2M	\$44.3M	\$60.3M	\$99.8M	\$157.3M

USD M	Y0 · 2026	Y2 · 2028	Y4 · 2030	Y5 · 2031	Y6 · 2032	Y8 · 2034	Y10 · 2036
Acquired revenue	\$0.0M	\$33.0M	\$64.8M	\$92.6M	\$103.7M	\$146.9M	\$184.3M
Total revenue	\$0.6M	\$36.0M	\$89.1M	\$136.9M	\$164.0M	\$246.7M	\$341.6M
Adjusted EBITDA	(\$1.3M)	\$2.7M	\$20.0M	\$41.1M	\$50.8M	\$81.4M	\$119.6M
Adjusted EBITDA margin	n/m	8%	23%	30%	31%	33%	35%
Cumulative acquisitions closed	0	2	4	5	5	6	6
Equity funding drawn in year	\$1.3M	\$32.6M	\$4.1M	\$1.7M	\$0.0M	\$0.0M	\$0.0M
Cumulative equity funding drawn	\$1.3M	\$34.5M	\$47.5M	\$49.3M	\$49.3M	\$49.3M	\$49.3M
eMACS units in service	2	5	11	14	17	23	29
Total FTE (implied)	11	242	561	844	1000	1484	2020

Base case Year 10 (2036): revenue \$342M (54% from acquired businesses), adjusted EBITDA \$120M at a 35% margin, 6 acquisitions, about 2,020 FTE, and 11.6% of the grown Year-10 SAM. Total equity drawn \$49M, all of it by 2031; the platform is self-funding from 2032. For comparison, the August package projected ~\$480M of Year-10 revenue from a modelled \$3.5M Year-0 base with seven acquisitions at 1.2x revenue and a hard-coded 20% margin.

6.2 Scenarios

Year 10 (2036)	Downside	Base	Upside
Revenue	\$115M	\$342M	\$694M
Adjusted EBITDA	\$35M	\$120M	\$278M
Adjusted EBITDA margin	30%	35%	40%
Acquisitions	4	6	7
Total equity funding 2026–36	\$48M	\$49M	\$50M
MOIC / IRR at Year-7 exit	4.2x / n/m	15.0x / 76%	37.6x / 112%
Share of Year-10 SAM	3.9%	11.6%	23.6%

Downside: every deal delayed a year and two dropped; 6% post-close growth; no margin uplift; slower fleet; 5% metals; 8x exit. It is a genuine failure case — the M&A programme stalls — and is shown because a private-equity reader will build it regardless. Upside: seven deals, 18% post-close growth, faster fleet, 15% metals, 12x exit. The single most load-bearing assumption in the plan is acquisition timing: deals made early compound for years; deals made late barely contribute.

6.3 Capital required and the ask

The base case draws \$49M of equity, front-loaded into Phases 2 and 3 (\$41M and \$6M), with only \$2.0M required in Phase 1. Gross uses over the decade: acquisition consideration \$83M, integration \$8.3M, fleet and maintenance capex \$50M, development and partnership programmes \$22M, working capital \$41M — with operating cash generation funding the balance. At the 2.0x-revenue stress price the acquisition programme alone would cost \$180M. Acquisition debt or seller paper would reduce the equity requirement; none is assumed.

The initial raise is confirmed at \$10M (Tom, 2 September 2026). This is not the full ten-year capital requirement above — it is sized against Phase 1 alone: \$1.97M fully funds the Phase 1 (2026–27)

operating build-out (fleet, platform team, certification layer, first allied-programme delivery), with the remaining \$8.03M held as a reserve toward first-acquisition diligence, negotiation and working capital. That reserve is deliberately not sized to close an acquisition — the first target alone is priced at ~\$17.8M and the first two combined at ~\$26.8M — so it funds getting to a signed LOI, not the close itself. In aggregate the initial raise is 23% of the combined Phase 1 + Phase 2 requirement: it buys the next 18–24 months, and a signed first-acquisition LOI is the milestone intended to size and trigger Round 2. Instrument (equity structure), valuation and target close are not yet set and are to be finalized with the lead investor; this memo and the accompanying comp/returns framework are offered as the reference point, not a quoted price.

7. Valuation and returns

7.1 Method

The August package applied a growth-stage go-public method: comp EV/EBITDA multiples on Year-10 EBITDA, discounted ten periods at 20%, against an illustrative \$150M deal EV. The teardown found the multiples stale, the 20% discount rate too low for the risk profile, the 20% Year-10 margin hard-coded, and the \$150M placeholder unbridged from the entry price. This package keeps the method as a cross-check — corrected — and leads with a private-equity returns case, which is the lens that fits a platform built by acquisition.

7.2 Comp universe (refreshed to late August 2026)

Reference	Data	Read
Cadre Holdings (NYSE: CDRE)	TTM revenue \$685.7M; EV \$1.62B; EV/revenue 2.36x; EV/EBITDA 15.3x; P/E 36.3x (31 Aug 2026)	Fundamentals anchor — EOD equipment, law-enforcement markets, 18 acquisitions incl. ICOR (Ottawa EOD robots, CAD 26M revenue, ~2.3x)
American Battery Technology (NASDAQ: ABAT)	TTM revenue \$16.3M; EV \$317M; EV/revenue 19.7x; net loss \$63.6M (26 Aug 2026)	Story anchor — critical-minerals recycling valued on narrative; not how this package prices itself
Public A&D government services	EV/EBITDA ~11.2x (KPMG Government Technology Solutions, end-2025)	Low end of the DFV comp range
Private A&D transactions	Median 18.96x TEV/EBITDA, 4.09x revenue (PCE, Q2 2026, 143 deals)	Upper reference only — large, de-risked targets
Founder-scale private market	GF Data 2025: sub-\$10M TEV 5.5x, \$10–25M 6.4x; Pepperdine 2025: 4.0–8.3x	Where the acquisition programme is priced (5.0–6.0x)

7.3 Returns case

Exit at Year 7 (2033) at 10x adjusted EBITDA of \$68.4M gives an enterprise value of \$684M; with closing cash and no debt, equity value against \$49M of equity drawn gives 15.0x and an IRR of 76% on the actual draw schedule. At Year 5 the same exit multiple gives 8.3x / 109% (capital is still being deployed); at Year 10, 29.3x / 54%. Exit-multiple sensitivity at Year 7: 6x → 9.5x; 8x → 12.3x; 12x → 17.8x.

7.4 The corrected discounted-future-value cross-check

Step	Value
Year-10 revenue (operating model)	\$341.6M
Year-10 adjusted EBITDA margin (operating model, not hard-coded)	35.0%
Year-10 adjusted EBITDA	\$119.6M
Comp multiple range (public A&D services 11.2x – Cadre 15.3x)	\$1,339M – \$1,829M
Discount rate / periods	25% / 10 (matched to the horizon)
Discounted firm value — midpoint	\$170M
Less equity deployed over the plan	\$49M
Net of capital deployed	\$121M
Same bridge, mismatched: 5 periods at 20%	\$637M
Same bridge at 30% / 10 periods	\$115M

Read: at a venture-grade discount rate over ten periods the cross-check now sits above the capital deployed — net of capital deployed of \$121M, versus a negative figure under the prior margin assumption. The swing follows mechanically from the higher Year-10 margin path (30% by Year 5, 35% by Year 10 — see Section 5.4) flowing through to Year-10 EBITDA and the comp-multiple bridge; it is not a re-estimate of the comps or the discount rate, both unchanged from the corrected August method. Because the bridge is this sensitive to the margin assumption, the returns case at a Year-7 exit, not the ten-year bridge, remains the primary lens presented in the deck. Neither figure is a deal price. The August package's illustrative \$150M deal EV and \$75M raise are withdrawn; no valuation is asserted.

8. Competitive position

The competitor set was rebuilt from each company's public materials. EnviroSafe Demil (Hooks, Texas; "America's clean alternative to open burning"; materials recovery; awardee on the 2021 Army demil contract) is the closest US comparable and the most important competitor to watch, with an undisclosed mobile lead time. Rheinmetall Expal operates fixed sites in Spain and Texas, has demilled more than 3.5 million munitions since 2017 and recycles more than 80% of material received — the August table's "no metals recovery" mark for Expal was wrong. GD-OTS runs demil at Carthage, Missouri (not Camden, Arkansas). Dynasafe supplies static detonation chambers, including transportable SDC 200/500 units, with the SDC 1000 stated to deploy in three weeks — the nearest closed-chamber comparable to a days-not-months claim. Day & Zimmermann operates more than 16,000 tons a year of demil capacity at Hawthorne under the new 20-year contract. El Dorado Engineering supplies a trailer-mounted flashing furnace for explosive-contaminated metal. General Atomics is a technology vendor, not an operator, and was removed from the table; the "no ITAR barrier" row was replaced with DFARS qualifying-country standing. What no incumbent combines is custody-mobile destruction, a police channel, and a mandate to assemble the founder-scale capability set.

9. Risks and mitigants

Risk	Why it matters	Mitigant / how the model treats it
M&A execution and pace	The plan is an investment-and-M&A story;	Named, ranked shortlist held for diligence;

Risk	Why it matters	Mitigant / how the model treats it
	late deals barely contribute	downside case delays every deal a year and drops two; capital drawn only as deals close
Acquisition pricing	August priced at 1.2x revenue; disclosed deals cleared 1.5–3.4x	Priced at 5.0–6.0x EBITDA with a 2.0x-revenue stress switch (\$180M vs \$83M)
Integration and margin	Founder-held specialists may under-perform post-close	Post-close growth 12% (not 15%); uplift phased over three years; integration cost 10% of price expensed
Regulatory classification	"Two-stage combustion" wording may class the unit as an incinerator under 40 CFR 260.10 rather than a Subpart X unit	Only the citable fact (closed chamber ≠ open burning) is claimed; counsel review before any further regulatory statement; certification track runs continuously
Recovery-rate and standards claims	97% recovery and CA EPA/NATO/EU standards are company figures	Labelled "company" throughout; no dollar-recovery tables; third-party verification to be commissioned and published
Title to recovered metal	In DoD demil services the Army sells the material (10 USC 7690)	Recovery modelled only on LE, commercial and allied work; 10% share (down from 14%); scenario range 5–15%
US market access	Canadian entity cannot hold US small-business status, ATF FEL or DDTC registration	DFARS 225.872 qualifying-country standing claimed instead; US entity established; set-aside claims removed
Market sizing	No analyst covers the category; several TAM layers are placeholders	Every layer labelled; SOM is the model's own revenue; diligence to harden allied budget and facility-engineering layers
Baseline scale	FY2025 combined revenue ~US\$0.6M and declining services revenue	Disclosed here; Year-0 anchored to actuals with a marked FY2026 placeholder; deck does not overstate current performance
Competitive response	EnviroSafe, Expal and Dynasafe can move toward mobile or recovery	Custody doctrine, police channel and roll-up mandate are the differentiators; EnviroSafe mobile lead time is the open question to close
Partnership placeholders	University licence in discussion; sensing partner unconfirmed	Shown as "name withheld" and "[TBC]"; no institution named until confirmed
Key-person and team	Deck team slide shows two confirmed people (founder + one strategic advisor); board not yet constituted	No fabricated names or roles used; board and further advisors to be added as those relationships close

10. The investment case

The need is sourced: a record rearmament with no end-of-life plan, a fixed-plant industry being regulated out, a US buyer that funds demil every year and holds 296,534 tons in the queue, and an episodic post-conflict layer already priced at \$27.6bn for one country. The foundation is real: twenty-five years of custody destruction, references on four continents, a US entity, and a doctrine no incumbent can retrofit. The platform is assembled rather than invented: six service lines, three engines, three phases, with the founder-scale capability priced at 4–7x EBITDA for now while strategics buy the large end. The plan is priced at real multiples from the actual baseline and lands at \$342M of Year-10 revenue on \$49M of equity, returning 15.0x / 76% at a Year-7 exit in the base case. The decision an investor is making is about capital deployment and M&A execution capability more than market size or story.

Recommended next steps before circulation

- Replace the FY2026 revenue placeholder (Baseline tab) with management's estimate; confirm units in service and the eMACS build cost per unit.

- Confirm the throughput figure to publish (the company's materials show 60 lb/hr and ~100 lb/cycle; the deck uses ~1,000 lb/day per unit) and commission third-party verification of the 97% recovery rate and the emissions-standard claims.
- Confirm or replace the sensing-partner placeholder; keep the dissolution-licence institution unnamed until signed.
- Team slide now shows Tom (Founder/CEO) and Joe Kasper (Strategic Advisor, confirmed 2 Sep 2026); add board members and any further advisors as those relationships close, and assemble contract references for the Australian, New Zealand and Saudi deployments under NDA.
- Environmental counsel review of the incinerator / Subpart X classification question before any regulatory statement beyond 40 CFR 260.10.
- Resolve the EnviroSafe Demil mobile lead-time question and monitor the 2026 Army CAD IDIQ award.
- Harden the assumed market layers (allied national demil budgets; sovereign facility engineering) through NSPA and national MoD interviews.
- Initial raise confirmed at \$10M. Instrument, valuation and target close remain to be set with the lead investor; the six-deal Round-2 raise size and use of proceeds follow once the first acquisition LOI is signed.

Appendix A — Changes from the August 2026 package

The full fact-check log (every claim, status, support and source link) is the Fact-Check Log tab of the model and the standalone document TWB_FactCheck_Log.md. The material changes:

Area	Change
Structure (Aug → Sep v1)	Deck reordered: Opportunity → Platform & Roadmap → Foundation → Investment case → Sources — leading with the scale of the opportunity and the platform vision, then grounding it in the operating proof before the numbers. Team slide deferred; capital-required and returns slides added; chemical-weapons line withdrawn.
Removed (Aug → Sep v1)	"\$808.8M IDIQ"; "\$3bn across 2026–27 demil awards"; "small-business set-aside for new entrants"; "2,600% antimony"; "\$1B Pentagon commitment"; "IACP Competitive Innovation Award" (finalist, not award); live-patent implication (patents expired, Crown-assigned); \$15bn TAM; \$3.5M modelled Year-0; 1.2x-revenue acquisition pricing; 20% discount rate; \$150M/\$75M illustrative deal; General Atomics as a competitor; the parent-company footer.
Corrected (Aug → Sep v1)	Ukraine \$29.8bn → \$27.6bn (RDNA5); State Dept CWD \$4bn → \$5.09bn/125+ countries; demil cost attribution GAO → NAS/Army; antimony "added 2025" → retained from 2022; "Ontario MoE" → MECP ECA / TSSA; GD-OTS location; Expal metals-recovery tick; Cadre and ABAT multiples refreshed to August 2026; EOD market \$9.0bn/4.5% → \$8.55bn/3.0%.
Added (Aug → Sep v1)	Verified demand signals (Army demil line, CAD IDIQ plan, Hawthorne contract, UN procurements, NATO requirements); critical-minerals primary data; competitor EnviroSafe Demil; scenario envelope; capital-by-phase; returns case; stockpile series reconciled (P-1 books and PD Demil forecast both dated).
Structure (Sep v1 → Sep v2, this package)	Revised against an external investor review (score 7.0/10, 2 Sep 2026). Section order: Foundation → Opportunity → Platform & Roadmap → Investment case (operating proof moved to slides 4–8, ahead of market sizing). eMACS and Unit Economics slides merged; the IP tile reframed from the now-expired 2004/2008 patents to process know-how, permits, references and custody data as the real moat. Platform & Roadmap compressed from five content slides to three (Assemble; Technology roadmap, now incorporating geography; Phase ladder) — the "One platform, four stages" and standalone Geography slides were folded in or cut as duplicative. Full competitive comparison table moved to a new appendix; the main deck now carries a simplified 2x2 positioning map. Closing slide rewritten to end on the specific round, milestones and investor decision rather than a general summary.
Added (Sep v1 → Sep v2, this package)	Traction slide (FY2023–25 actuals — revenue, EBITDA, EBITDA margin, net income, balance sheet — already in this memo, now surfaced in the deck; fleet-in-service count still withheld pending confirmation). Team slide (Tom, Founder/CEO; Joe Kasper, Strategic Advisor — the only two people named; advisory relationship confirmed by Tom, 2 Sep 2026). Dedicated Ask slide: the \$10M initial raise, use of proceeds, and milestones funded. De-risking roadmap slide: the specific verification, reference-call, LOI and disclosure milestones that trigger Round 2.
Financing (Sep v1 → Sep v2, this package)	The initial raise is now confirmed at \$10M (Tom, 2 Sep 2026) — see section 6.3. This is new, company-provided information; no other TWB operating data was added to the deck beyond what this memo and model already carried as verified, at Tom's explicit instruction to add no further unconfirmed company data to the investor deck for this round.
Open items	FY2026 revenue placeholder; units in service and unit build cost; throughput figure to publish; sensing-partner name; third-party verification of recovery/emissions claims; raise instrument, valuation and target close; board composition beyond the two team-slide members; the brand logo in the deck is the master PNG — confirm it renders correctly on the presenting machine.

Appendix B — Principal sources

SIPRI, Trends in World Military Expenditure 2025 (Apr 2026) — https://www.sipri.org/sites/default/files/2026-04/2604_milex_2025.pdf

UCDP / Uppsala University press release (9 Jun 2026) — <https://www.uu.se/en/press/press-releases/2026/2026-06-09-ucdp-record-number-of-conflicts-between-states>

US Army FY2027 Budget Estimates — Procurement of Ammunition (Apr 2026) — https://www.asafm.army.mil/Portals/72/Documents/BudgetMaterial/2027/Discretionary%20Budget/Procurement/Procurement_of_Ammunition.pdf

PD Demil, NDIA Global Demil Symposium (1 Oct 2025) — https://www.ndia.org/-/media/sites/ndia/meetings-and-events/proceedings/2025/FFC/October%201/Hrycak_Orest_2025%20FFC_1%20October%202025.pdf

World Bank / Ukraine RDNA5 (23 Feb 2026) —

<https://www.worldbank.org/en/news/press-release/2026/02/23/updated-ukraine-recovery-and-reconstruction-needs-assessment-released>

Federal Register — EPA proposed OB/OD rule, 89 FR 19952 —

<https://www.federalregister.gov/documents/2024/03/20/2024-05088/revisions-to-standards-for-the-open-burningopen-detonation-of-waste-explosives>

Federal Register — Final 2025 List of Critical Minerals —

<https://www.federalregister.gov/documents/2025/11/07/2025-19813/final-2025-list-of-critical-minerals>

USGS Mineral Commodity Summaries 2026 — Antimony — <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-antimony.pdf>

National Academies, Alternatives for the Demilitarization of Conventional Munitions (2019) —

<https://www.nationalacademies.org/read/25140/chapter/4>

DoD Contracts for 9 Jun 2026 (Hawthorne) — https://www.globalsecurity.org/military/library/news/2026/06/dod-contracts_4512718.htm

UNGM EOIUNPD23750 (Jul 2025) — <https://www.ungm.org/Public/Notice/271658>

Google Patents US6758151B2 — <https://patents.google.com/patent/US6758151B2/en>

Officer.com — eMACS listing (2011) and 2011 Innovation Awards results — <https://www.officer.com/tactical/firearm-accessories/product/10446308/twb-designs-inc-emacs-ammo-burner>

StockAnalysis — Cadre Holdings statistics (31 Aug 2026) — <https://stockanalysis.com/stocks/cdre/statistics/>

PCE Aerospace, Defense & Government M&A update (Q2 2026) —

<https://www.pcecompanies.com/resources/industry-reports/aerospace-government-ma-update>

GF Data / Pepperdine lower-middle-market multiples (2025) — <https://capitalpad.com/lower-middle-market-ebitda-multiples/>

Bank of Canada annual average exchange rates — <https://www.bankofcanada.ca/rates/exchange/annual-average-exchange-rates/>